

About This Document

The technical interview for a Data Engineer Trainee covers SQL, databases, data warehousing, ETL/pipelines, big data and the cloud/data-platform stack (Snowflake, Azure) BI3 works with.

This document has topic-wise MCQs (correct answer highlighted) followed by short-answer interview questions with model answers.

SQL and Databases

Q1. Which SQL clause is used to filter rows AFTER aggregation (on grouped results)?

(A) WHERE

(B) HAVING ✓

(C) GROUP BY

(D) ORDER BY

Answer: (B) HAVING

Q2. Which JOIN returns all rows from the left table and matching rows from the right, with NULLs where there is no match?

(A) INNER JOIN

(B) LEFT JOIN ✓

(C) CROSS JOIN

(D) SELF JOIN

Answer: (B) LEFT JOIN

Q3. A PRIMARY KEY constraint ensures that a column's values are:

(A) Sorted

(B) Unique and not null ✓

(C) Indexed only

(D) Foreign

Answer: (B) Unique and not null

Q4. Which normal form removes partial dependency of non-key attributes on part of a composite key?

(A) 1NF

(B) 2NF ✓

(C) 3NF

(D) BCNF

Answer: (B) 2NF

Q5. Which SQL statement is used to remove all rows quickly without logging individual row deletions and cannot be rolled back easily?

(A) DELETE

(B) TRUNCATE ✓

(C) DROP

(D) REMOVE

Answer: (B) TRUNCATE

Q6. Which of the following is an aggregate function?

(A) UPPER

(B) LENGTH

(C) AVG ✓

(D) SUBSTR

Answer: (C) AVG

Data Warehousing and Modeling

Q1. In dimensional modeling, the central table containing measures/metrics is called the:

(A) Dimension table

(B) Fact table ✓

(C) Lookup table

(D) Bridge table

Answer: (B) Fact table

Q2. A schema with a central fact table connected directly to denormalized dimension tables is called a:

(A) Snowflake schema

(B) Star schema ✓

(C) Galaxy schema

(D) Network schema

Answer: (B) Star schema

Q3. A snowflake schema differs from a star schema mainly because its dimension tables are:

(A) Denormalized

(B) Normalized into sub-dimensions ✓

(C) Removed

(D) Combined

Answer: (B) Normalized into sub-dimensions

Q4. OLAP systems are primarily optimized for:

(A) High-volume transactional writes

(B) Analytical queries and aggregation ✓

(C) Row-level locking

(D) Payment processing

Answer: (B) Analytical queries and aggregation

Q5. A slowly changing dimension (SCD) Type 2 handles changes by:

(A) Overwriting the old value

(B) Adding a new row with versioning ✓

(C) Deleting the record

(D) Ignoring the change

Answer: (B) Adding a new row with versioning

Q6. A data warehouse typically stores data that is:

(A) Real-time only

(B) Subject-oriented, integrated and historical ✓

(C) Unstructured only

(D) Temporary

Answer: (B) Subject-oriented, integrated and historical

ETL, Pipelines and Big Data

Q1. In the acronym ETL, the letters stand for:

(A) Extract, Transform, Load ✓

(B) Export, Transfer, Link

(C) Extract, Transfer, Log

(D) Evaluate, Transform, Load

Answer: (A) Extract, Transform, Load

Q2. The key difference between ETL and ELT is that in ELT:

(A) Data is never transformed

(B) Transformation happens after loading into the target system ✓

(C) Data is only extracted

(D) Loading is skipped

Answer: (B) Transformation happens after loading into the target system

Q3. Which processing model handles data in real time as it arrives?

(A) Batch processing

(B) Stream processing ✓

(C) Manual processing

(D) Nightly processing

Answer: (B) Stream processing

Q4. Apache Spark primarily processes data using:

(A) Disk-only MapReduce

(B) In-memory distributed computing ✓

(C) Single-threaded loops

(D) Stored procedures

Answer: (B) In-memory distributed computing

Q5. In a data pipeline, idempotency means that re-running the same step:

(A) Doubles the data

(B) Produces the same result without duplicates ✓

(C) Fails every time

(D) Deletes the source

Answer: (B) Produces the same result without duplicates

Q6. Partitioning a large table primarily helps to:

(A) Increase storage cost

(B) Improve query performance by pruning data ✓

(C) Remove all indexes

(D) Prevent joins

Answer: (B) Improve query performance by pruning data

Cloud and Snowflake

Q1. Snowflake separates storage and compute so that:

(A) They must scale together

(B) They can scale independently ✓

(C) Compute is disabled

(D) Storage is limited

Answer: (B) They can scale independently

Q2. In Snowflake, the compute resource that runs queries is called a:

(A) Stage

(B) Virtual warehouse ✓

(C) Micro-partition

(D) Role

Answer: (B) Virtual warehouse

Q3. A data lake typically stores:

(A) Only structured relational data

(B) Raw data in many formats (structured, semi-structured, unstructured) ✓

(C) Only aggregated reports

(D) Only images

Answer: (B) Raw data in many formats (structured, semi-structured, unstructured)

Q4. Which file format is columnar and commonly used in big-data analytics?

(A) CSV

(B) Parquet ✓

(C) TXT

(D) XML

Answer: (B) Parquet

Q5. Object storage services like Amazon S3 or Azure Blob are best described as:

(A) Relational databases

(B) Scalable storage for files/objects ✓

(C) Message queues

(D) Compute clusters

Answer: (B) Scalable storage for files/objects

Python for Data Engineering

Q1. Which Python library is most commonly used for tabular data manipulation?

(A) NumPy

(B) pandas ✓

(C) matplotlib

(D) requests

Answer: (B) pandas

Q2. In pandas, which method removes rows with missing (NaN) values?

(A) fillna()

(B) dropna() ✓

(C) isnull()

(D) merge()

Answer: (B) dropna()

Q3. Which pandas operation combines two DataFrames based on a key column, similar to a SQL join?

(A) concat()

(B) merge() ✓

(C) append()

(D) pivot()

Answer: (B) merge()

Q4. Which data structure in Python stores unique, unordered elements?

(A) list

(B) tuple

(C) set ✓

(D) dict

Answer: (C) set

Q5. In pandas, groupby() is typically followed by:

(A) A sort only

(B) An aggregation such as sum() or mean() ✓

(C) Deletion

(D) Nothing

Answer: (B) An aggregation such as sum() or mean()

Data Engineering Interview - Short Answers

Q1. What is the difference between a data warehouse and a data lake?

Answer: A data warehouse stores structured, processed, schema-on-write data optimised for analytics and reporting; a data lake stores raw data of any format (schema-on-read) and is cheaper and more flexible but needs more processing before analysis.

Q2. Explain the difference between OLTP and OLAP.

Answer: OLTP systems handle many short transactional read/write operations (e.g. order entry) with normalized schemas; OLAP systems handle complex analytical queries over large historical data with denormalized, aggregation-friendly schemas.

Q3. What is the difference between DELETE, TRUNCATE and DROP in SQL?

Answer: DELETE removes selected rows and can be rolled back and filtered with WHERE; TRUNCATE removes all rows quickly with minimal logging and resets the table; DROP removes the entire table structure and its data.

Q4. What is a primary key versus a foreign key?

Answer: A primary key uniquely identifies each row in a table and cannot be null; a foreign key is a column that references the primary key of another table to enforce referential integrity between them.

Q5. What is normalization and why is it used?

Answer: Normalization organises data into related tables to remove redundancy and update anomalies (1NF, 2NF, 3NF). It improves data integrity, though heavy analytics workloads sometimes denormalize for speed.

Q6. What is the difference between batch and stream processing?

Answer: Batch processing handles large volumes of data collected over a period (e.g. nightly jobs); stream processing handles data continuously in near real time as each event arrives (e.g. Kafka + Spark Streaming).

Q7. What is a star schema?

Answer: A star schema has a central fact table holding measures and foreign keys, surrounded by denormalized dimension tables. It is simple and fast for analytical queries and is widely used in data warehouses.

Q8. How would you optimize a slow SQL query?

Answer: Add appropriate indexes, select only needed columns, filter early, avoid unnecessary joins and subqueries, use partitioning, examine the query execution plan, and precompute or materialise heavy aggregations.

Data Engineering Concepts

Applied Role(s):

Data Engineer - Trainee

Online Assessment Preparation Material

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Compiled from verified previous-year patterns, community-reported questions and research-backed company-style practice.

Table of Contents

About This Document	1
SQL and Databases	1
Data Warehousing and Modeling	2
ETL, Pipelines and Big Data	3
Cloud and Snowflake	4
Python for Data Engineering	5
Data Engineering Interview - Short Answers	6